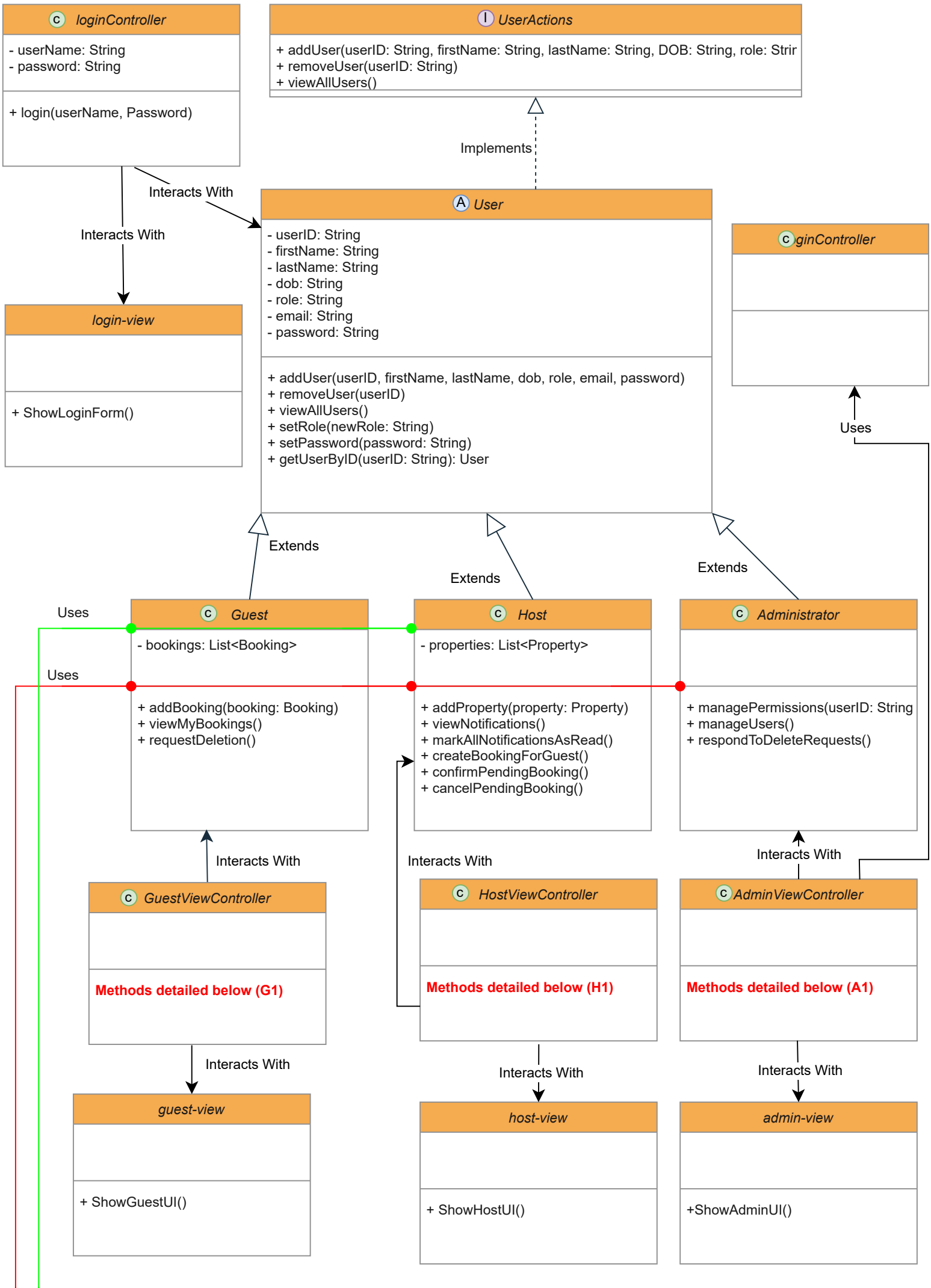
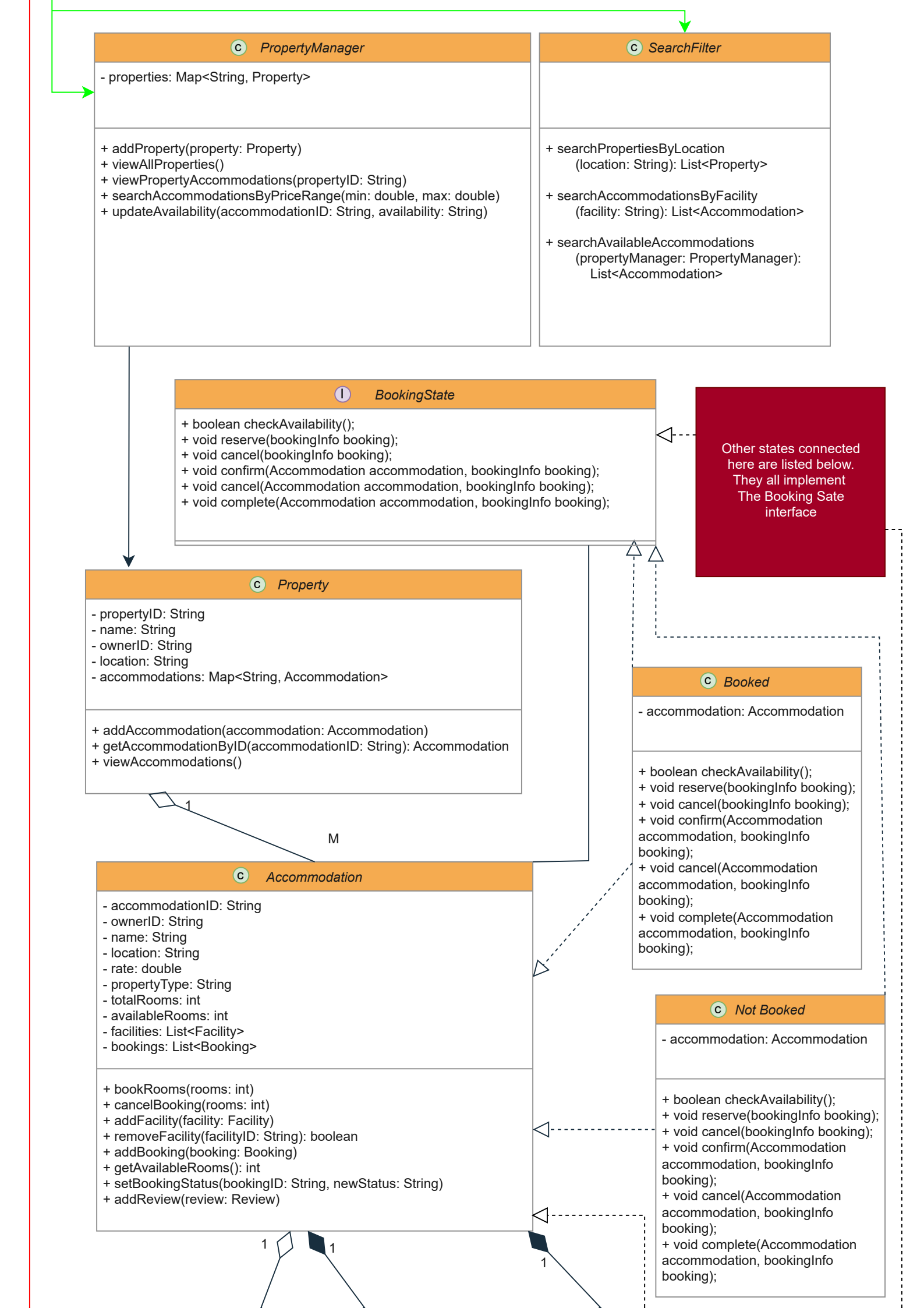
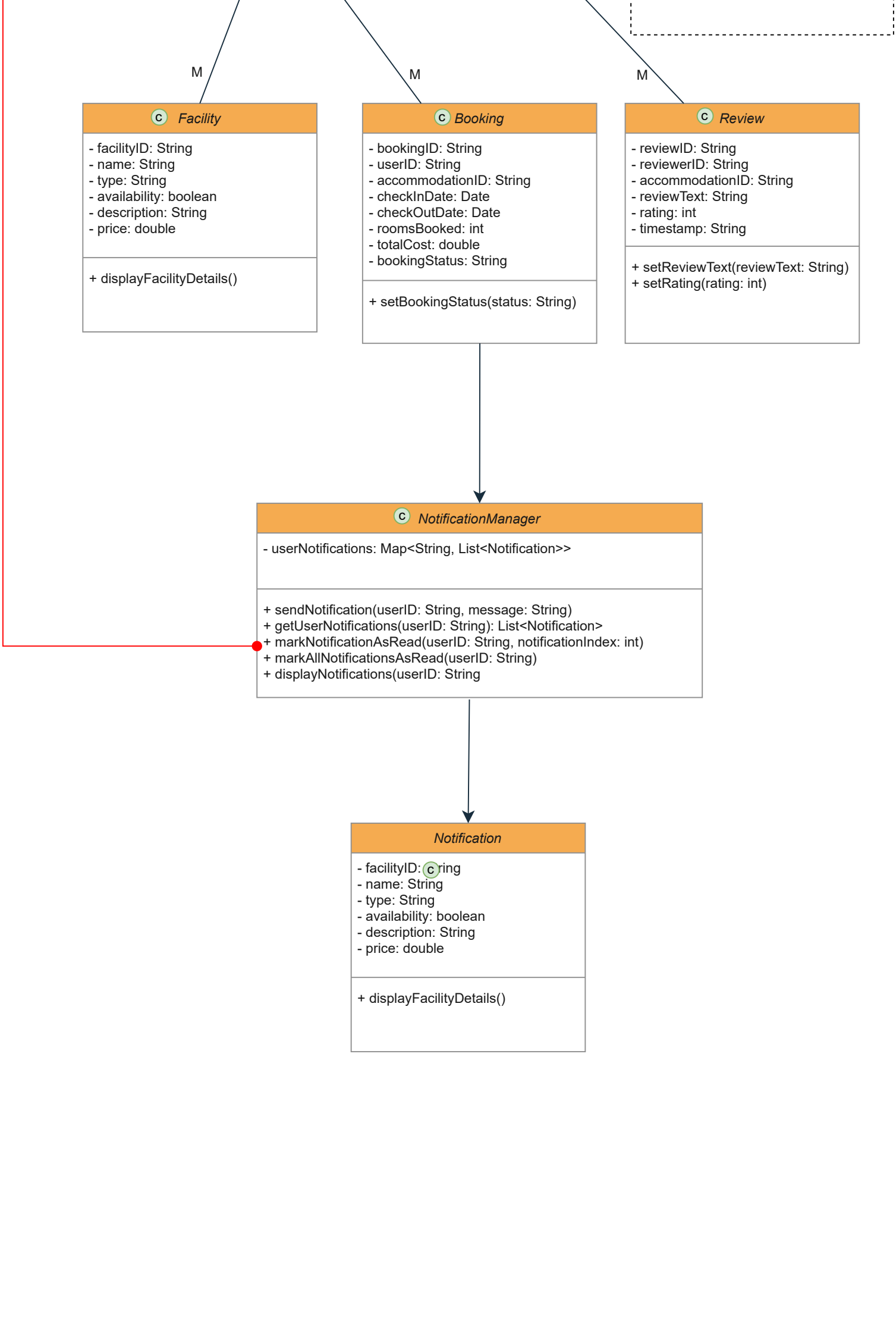

loginController
- userName: String
- password: String
+ login(userName, Password)
UserActions
+ addUser(userID: String, firstName: String, lastName: String, DOB: String, role: Strir
+ removeUser(userID: String)
+ viewAllUsers()
Implements
Interacts With
Interacts With
User
- userID: String
- firstName: String
- lastName: String
- dob: String
- role: String
- email: String
- password: String
+ addUser(userID, firstName, lastName, dob, role, email, password)
+ removeUser(userID)
+ viewAllUsers()
+ setRole(newRole: String)
+ setPassword(password: String)
+ getUserByID(userID: String): User
login-view
+ ShowLoginForm()
ginController
Uses
Extends
Extends
Extends
Uses
Uses
Guest
- bookings: List<Booking>
+ addBooking(booking: Booking)
+ viewMyBookings()
+ requestDeletion()
Host
- properties: List<Property>
+ addProperty(property: Property)
+ viewNotifications()
+ markAllNotificationsAsRead()
+ createBookingForGuest()
+ confirmPendingBooking()
+ cancelPendingBooking()
Administrator
+ managePermissions(userID: String
+ manageUsers()
+ respondToDeleteRequests()
Interacts With
Interacts With
Interacts With
GuestViewController
Methods detailed below (G1)
HostViewController
Methods detailed below (H1)
AdminViewController
Methods detailed below (A1)
Interacts With
Interacts With
Interacts With
guest-view
+ ShowGuestUI()
host-view
+ ShowHostUI()
admin-view
+ShowAdminUI()


PropertyManager
- properties: Map<String, Property>
+ addProperty(property: Property)
+ viewAllProperties()
+ viewPropertyAccommodations(propertyID: String)
+ searchAccommodationsByPriceRange(min: double, max: double)
+ updateAvailability(accommodationID: String, availability: String)
SearchFilter
+ searchPropertiesByLocation (location: String): List<Property>
+ searchAccommodationsByFacility (facility: String): List<Accommodation>
+ searchAvailableAccommodations (propertyManager: PropertyManager): List<Accommodation>
BookingState
+ boolean checkAvailability();
+ void reserve(bookingInfo booking);
+ void cancel(bookingInfo booking);
+ void confirm(Accommodation accommodation, bookingInfo booking);
+ void cancel(Accommodation accommodation, bookingInfo booking);
+ void complete(Accommodation accommodation, bookingInfo booking);
Other states connected here are listed below. They all implement The Booking Sate interface
Property
- propertyID: String
- name: String
- ownerID: String
- location: String
- accommodations: Map<String, Accommodation>
+ addAccommodation(accommodation: Accommodation)
+ getAccommodationByID(accommodationID: String): Accommodation
+ viewAccommodations()
1
M
Booked
- accommodation: Accommodation
+ boolean checkAvailability();
+ void reserve(bookingInfo booking);
+ void cancel(bookingInfo booking);
+ void confirm(Accommodation accommodation, bookingInfo booking);
+ void cancel(Accommodation accommodation, bookingInfo booking);
+ void complete(Accommodation accommodation, bookingInfo booking);
Accommodation
- accommodationID: String
- ownerID: String
- name: String
- location: String
- rate: double
- propertyType: String
- totalRooms: int
- availableRooms: int
- facilities: List<Facility>
- bookings: List<Booking>
+ bookRooms(rooms: int)
+ cancelBooking(rooms: int)
+ addFacility(facility: Facility)
+ removeFacility(facilityID: String): boolean
+ addBooking(booking: Booking)
+ getAvailableRooms(): int
+ setBookingStatus(bookingID: String, newStatus: String)
+ addReview(review: Review)
Not Booked
- accommodation: Accommodation
+ boolean checkAvailability();
+ void reserve(bookingInfo booking);
+ void cancel(bookingInfo booking);
+ void confirm(Accommodation accommodation, bookingInfo booking);
+ void cancel(Accommodation accommodation, bookingInfo booking);
+ void complete(Accommodation accommodation, bookingInfo booking);
1
1
1


M
M
M
Facility
- facilityID: String
- name: String
- type: String
- availability: boolean
- description: String
- price: double
+ displayFacilityDetails()
Booking
- bookingID: String
- userID: String
- accommodationID: String
- checkInDate: Date
- checkOutDate: Date
- roomsBooked: int
- totalCost: double
- bookingStatus: String
+ setBookingStatus(status: String)
Review
- reviewID: String
- reviewerID: String
- accommodationID: String
- reviewText: String
- rating: int
- timestamp: String
+ setReviewText(reviewText: String)
+ setRating(rating: int)
NotificationManager
- userNotifications: Map<String, List<Notification>>
+ sendNotification(userID: String, message: String)
+ getUserNotifications(userID: String): List<Notification>
+ markNotificationAsRead(userID: String, notificationIndex: int)
+ markAllNotificationsAsRead(userID: String)
+ displayNotifications(userID: String
Notification
- facilityID: String
- name: String
- type: String
- availability: boolean
- description: String
- price: double
+ displayFacilityDetails()

Other States

PendingState

- accommodation: Accommodation

+ boolean checkAvailability();
+ void reserve(bookingInfo booking);
+ void cancel(bookingInfo booking);
+ void confirm(Accommodation accommodation, bookingInfo booking);
+ void cancel(Accommodation accommodation, bookingInfo booking);
+ void complete(Accommodation accommodation, bookingInfo booking);

ConfirmedState

- accommodation: Accommodation

+ boolean checkAvailability();
+ void reserve(bookingInfo booking);
+ void cancel(bookingInfo booking);
+ void confirm(Accommodation accommodation, bookingInfo booking);
+ void cancel(Accommodation accommodation, bookingInfo booking);
+ void complete(Accommodation accommodation, bookingInfo booking);

CanceledState

+ boolean checkAvailability();
+ void reserve(bookingInfo booking);
+ void cancel(bookingInfo booking);
+ void confirm(Accommodation accommodation, bookingInfo booking);
+ void cancel(Accommodation accommodation, bookingInfo booking);
+ void complete(Accommodation accommodation, bookingInfo booking);

CompletedState

+ boolean checkAvailability();
+ void reserve(bookingInfo booking);
+ void cancel(bookingInfo booking);
+ void confirm(Accommodation accommodation, bookingInfo booking);
+ void cancel(Accommodation accommodation, bookingInfo booking);
+ void complete(Accommodation accommodation, bookingInfo booking);

G1,H1,A1 Details

A1

- populateRoleChoiceBox(): Void
- showAlert(String title, String message, Alert.AlertType alertType): Void
- deleteUserRequest(): Void
- updateNewUser(): Void
- addNewUser(): Void
- clearAddUserFields(): Void
- isValidDate(String date): Boolean
- isValidEmail(String email): Boolean
- populatePerformanceChart(): Void
+ adminSignOut(): Void
+ populateDeleteRequests(): Void
+ populateUserIDs(): Void
+ populateOverviewChart(): Void
+ initialize(): Void

G1

- showAlert(String title, String message, Alert.AlertType alertType): Void

+ cancelBooking(): Void
+ GuestSignOut(): Void
+ makeBooking(): Void
+ makeDeleteRequest(): Void
+ markAsRead(): Void
+ populateNotifications(User activeUser): Void
+ updatePerformanceTracker(): Void

+ viewAccommodations(PropertyManagerCommand propertyManager): Void

+ viewBookings(User activeUser): Void
+ writeReview(): Void

H1

+ hostSignOut(): Void
+ makeBooking(): Void
+ updateAvailability(): Void
+ markAsRead(): Void
+ completeCurrentBooking(): Void
+ confirmPendingBooking(): Void
+ cancelPendingBooking(): Void
+ cancelCurrentBooking(): Void
+ manageFacility(): Void
+ makeDeleteRequest(): Void
+ updatePerformanceTracker(): Void

+ populateCompletedBookings(User activeUser, PropertyManagerCommand propertyManager): Void

+ populateMyAccommodations(User activeUser, PropertyManagerCommand propertyManager): Void

+ viewBookings(User activeUser, PropertyManagerCommand propertyManager): Void

+ listNewAccommodation(User activeUser): Void

+ populateNotifications(User activeUser): Void

+ populateReviews(User activeUser, PropertyManagerCommand propertyManager): Void

+ populatePendingBookings(User activeUser, PropertyManagerCommand propertyManager): Void